

Problem

How much energy does a 10-hp motor deliver in 30 minutes? Assume that 1 horsepower = 746 W.

Step-by-step solution

Step 1 of 2

Determine the energy delivered by the motor E .

Consider the motor power $p = 10 \text{ hp}$.

Consider,

$$1\text{-hp} = 746 \text{ W}$$

Calculate the power delivered by 10 hp motor.

$$\begin{aligned} p &= 10\text{-hp} \\ &= 7460 \text{ W} \end{aligned}$$

Step 2 of 2

Consider the time period is 30 min.

Convert minutes into hours.

$$\begin{aligned} t &= 30 \text{ min} \\ &= \frac{1}{2} \text{ hr} \end{aligned}$$

Consider the equation for energy.

$$E = pt$$

Replace 7460 W for p and $\frac{1}{2} \text{ hr}$ for t in equation.

$$\begin{aligned} E &= (7460) \left(\frac{1}{2} \right) \\ &= 3730 \text{ Wh} \end{aligned}$$

Therefore, the energy delivered by the 10 hp motor in 30 min is 3730 Wh.